

FRAUDULENT CLAIM DETECTION CASE STUDY REPORT

Problem Statement -

Global Insure, a leading insurance company, processes thousands of claims annually. However, a significant percentage of these claims turn out to be fraudulent, resulting in considerable financial losses. The company's current process for identifying fraudulent claims involves manual inspections, which is time-consuming and inefficient. Fraudulent claims are often detected too late in the process, after the company has already paid out significant amounts. Global Insure wants to improve its fraud detection process using data-driven insights to classify claims as fraudulent or legitimate early in the approval process. This would minimise financial losses and optimise the overall claims handling process.

Business Objective -

Global Insure wants to build a model to classify insurance claims as either fraudulent or legitimate based on historical claim details and customer profiles. By using features like claim amounts, customer profiles and claim types, the company aims to predict which claims are likely to be fraudulent before they are approved.

Based on this assignment, you have to answer the following questions:

- How can we analyse historical claim data to detect patterns that indicate fraudulent claims?
- Which features are most predictive of fraudulent behaviour?
- Can we predict the likelihood of fraud for an incoming claim, based on past data?
- What insights can be drawn from the model that can help in improving the fraud detection process?

Methodology -

1. Data Preparation
 - a. Import necessary libraries and load the data
2. Data Cleaning
 - a. Handling null values
 - b. Handling redundant features
 - c. Fix data types
3. Train-Validation Split
 - a. Define feature and target(s) variables
 - b. Split data into training and validation sets
4. EDA on Training Data
 - a. Perform univariate analysis
 - b. Perform correlation analysis
 - c. Check class balance
 - d. Perform bivariate analysis
5. Feature Engineering
 - a. Perform resampling
 - b. New feature creation
 - c. Handle redundant columns
 - d. Combine values in categorical columns
 - e. Dummy variable creation
 - f. Feature rescaling
6. Model Building
 - a. Feature selection
 - b. Building logistic regression model
 - c. Find the optimal cutoff
 - d. Building random forest model
 - e. Hyperparameter tuning
7. Predictions and Model Evaluation

- Make predictions over validation data using the logistic regression model
- Make predictions over validation data using the random forest model

Visualisations -

1. Logistic Regression

Optimization terminated successfully.
Current function value: 0.396638
Iterations: 7

Logit Regression Results						
Dep. Variable:	fraud_reported	No. Observations:	331			
Model:	logit	DF residuals:	302			
Method:	NLE	DF Model:	28			
Date:	Tue, 06 May 2025	Pseudo R-squ.:	0.2736			
Time:	14:54:25	Log-Likelihood:	-131.28			
Converged:	TRUE	LL-Null:	-180.73			
Covariance Type:	nonrobust	LLR p-value:	7.653e-10			
	coef	std err	z	P> z	[0.025	0.975]
const	-1.6662	1.511	-1.103	0.270	-4.627	1.294
log_vehicle_claim	0.4556	0.666	0.684	0.494	-0.849	1.760
policy_csl_250/500	0.2123	0.383	0.555	0.579	-0.538	0.962
policy_csl_500/1000	0.5706	0.411	1.388	0.165	-0.235	1.377
insured_relationship_not-in-family	0.3705	0.406	0.367	0.714	-0.775	1.132
insured_relationship_other-relative	0.7613	0.469	1.624	0.104	-0.158	1.682
insured_relationship_unmarried	1.1184	0.695	2.258	0.024	0.147	2.089
insured_relationship_wife	0.9337	0.555	1.672	0.095	-0.161	2.028
collision_type_Front Collision	1.7383	1.826	0.945	0.345	-1.840	5.388
collision_type_Rear Collision	2.8068	1.733	1.168	0.242	-1.371	5.424
collision_type_Side Collision	1.2895	1.733	0.744	0.457	-2.187	4.686
incident_severity_Minor Damage	-2.2449	0.428	-5.239	0.000	-3.085	-1.405
incident_severity_Total Loss	-2.2002	0.431	-5.290	0.000	-3.125	-1.435
authorities_contacted_Fire	-0.2246	0.507	-0.443	0.655	-1.119	0.770
authorities_contacted_Other	0.4174	0.498	0.839	0.401	-0.558	1.393
authorities_contacted_Police	0.2559	0.455	0.559	0.575	-0.642	1.156
incident_state_NV	-1.0426	0.460	-2.265	0.023	-1.949	-0.142
incident_state_SC	-0.1158	0.412	-0.281	0.779	-0.923	0.692
incident_state_WV	-0.0809	0.514	-0.154	0.887	-1.088	0.927
incident_city_Columbus	-0.3931	0.510	-0.771	0.441	-1.392	0.605
incident_city_Northbend	-0.4723	0.457	-1.034	0.301	-1.367	0.423
incident_city_Riverwood	-0.0546	0.487	-0.112	0.911	-1.009	0.900
incident_city_Springfield	-0.5151	0.522	-0.969	0.333	-1.550	0.527
property_damage_NO	-0.4544	0.396	-1.151	0.251	-1.241	0.333
property_damage_YES	-0.2671	0.390	-0.670	0.503	-1.042	0.514
police_report_available_NO	0.4324	0.389	1.138	0.255	-0.312	1.177
police_report_available_YES	-0.3864	0.430	-0.898	0.369	-1.238	0.457
auto_make_Ford	0.5085	0.565	1.748	0.080	-0.120	2.497
auto_make_Mercedes	0.5487	0.625	1.505	0.131	-0.104	2.181

	feature	VIF
0	const	69.566356
1	log_vehicle_claim	13.792337
2	policy_csl_250/500	1.392226
3	policy_csl_500/1000	1.457725
4	insured_relationship_not-in-family	1.308258
5	insured_relationship_other-relative	1.393616
6	insured_relationship_unmarried	1.278336
7	insured_relationship_wife	1.277173
8	collision_type_Front Collision	15.498072
9	collision_type_Rear Collision	19.197976
10	collision_type_Side Collision	18.268946
11	incident_severity_Minor Damage	1.363464
12	incident_severity_Total Loss	1.462363
13	authorities_contacted_Fire	1.760876
14	authorities_contacted_Other	1.637703
15	authorities_contacted_Police	1.581578
16	incident_state_NV	1.612650
17	incident_state_SC	1.488002
18	incident_state_WV	1.443963
19	incident_city_Columbus	1.280845
20	incident_city_Northbend	1.234612
21	incident_city_Riverwood	1.251408
22	incident_city_Springfield	1.219598
23	property_damage_NO	1.435174
24	property_damage_YES	1.401899
25	police_report_available_NO	1.363204
26	police_report_available_YES	1.316319
27	auto_make_Ford	1.073301
28	auto_make_Mercedes	1.095583

Predicted Classification:

	Actual	Predicted_Prob	
0	0	0.543954	
1	1	0.625232	
2	0	0.020952	
3	1	0.004596	
4	0	0.058728	
	Actual	Predicted_Prob	Predicted_Class
0	0	0.543954	1
1	1	0.625232	1
2	0	0.020952	0
3	1	0.004596	1
4	0	0.058728	0

Model Accuracy – 0.8128

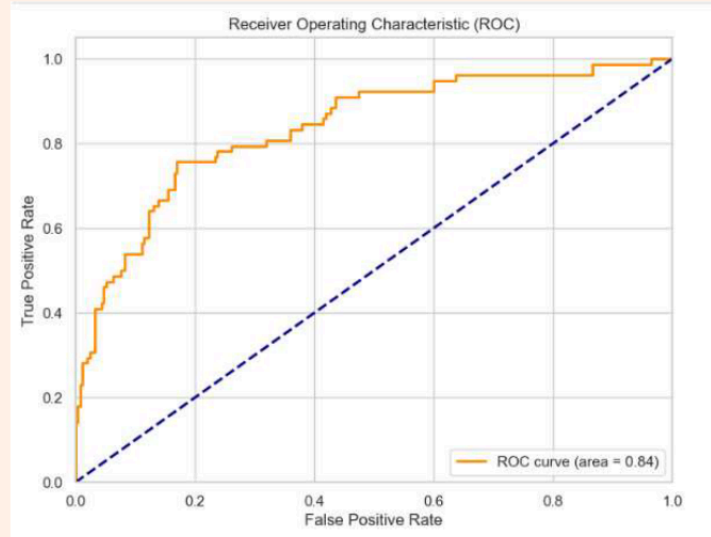
Confusion Metrics:

Confusion Matrix:

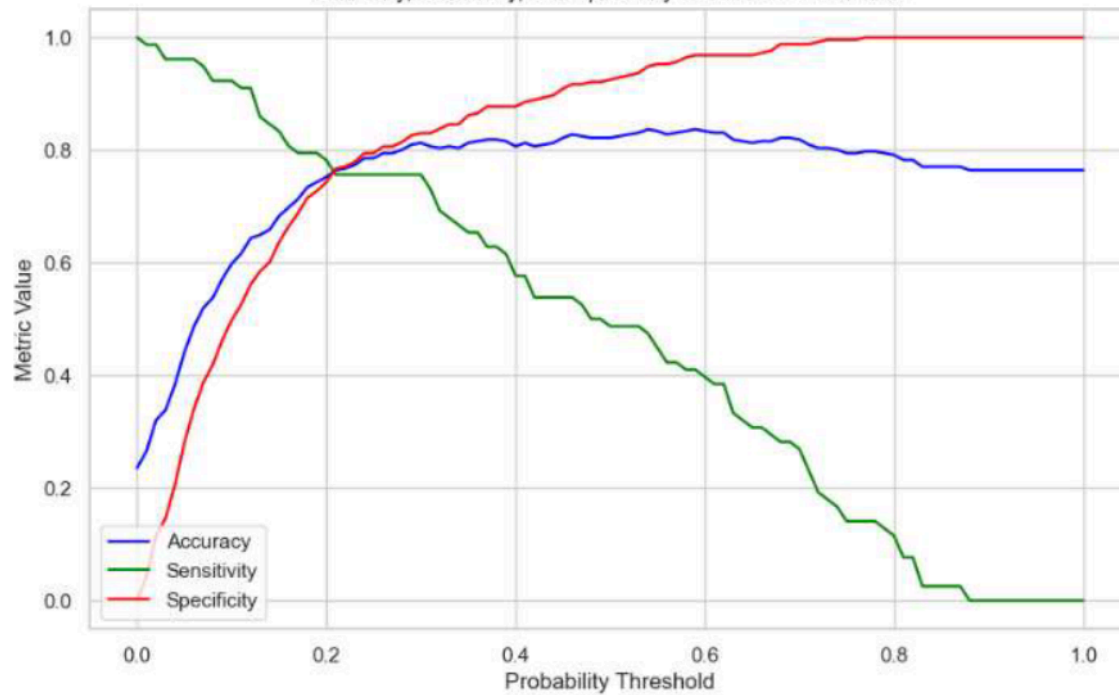
```
[[234 19]
 [ 40 38]]
```

Sensitivity: 0.4872
Specificity: 0.9249
Precision: 0.6667
Recall: 0.4872
F1 Score: 0.5630

ROC Curve



Accuracy, Sensitivity, and Specificity at Different Thresholds



Final Prediction:

	Actual	Predicted_Prob	Predicted_Class	Final_Prediction
0	0	0.543954	1	1
1	1	0.625232	1	1
2	0	0.828952	0	0
3	1	0.804596	1	1
4	0	0.858728	0	0

Final Accuracy – 0.8218

Confusion Metrics:

Final Confusion Matrix:

```
[[234  19]
 [ 40  38]]
```

Final Sensitivity: 0.4872

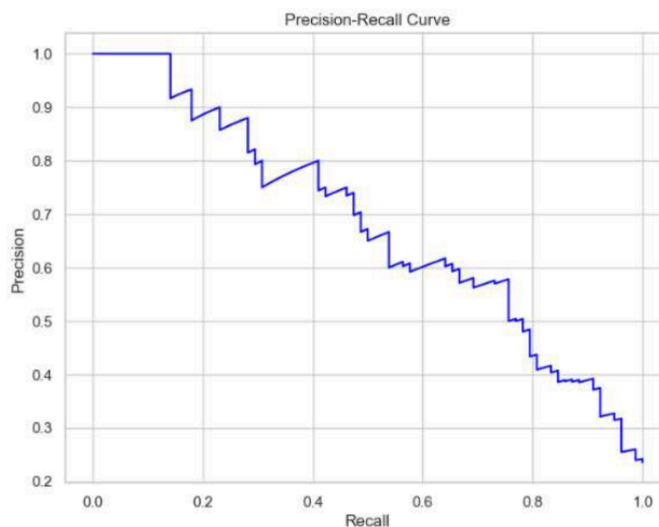
Final Specificity: 0.9249

Final Precision: 0.6667

Final Recall: 0.4872

Final F1 Score: 0.5630

Precision Recall Curve



2. Random Forest

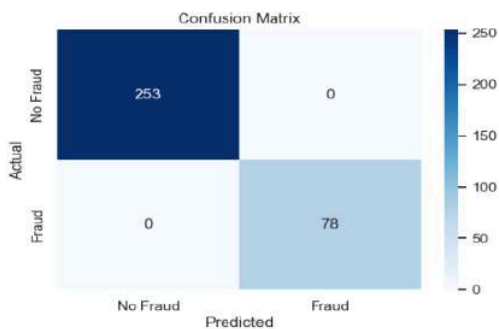
Final Accuracy – 1

Cross Validation Score

Cross-validation scores: [0.76119403 0.78787879 0.81818182 0.78787879 0.75757576]

Mean cross-validation score: 0.7825418362731795

Confusion Metrics:



Sensitivity (Recall): 1.0

Specificity: 1.0

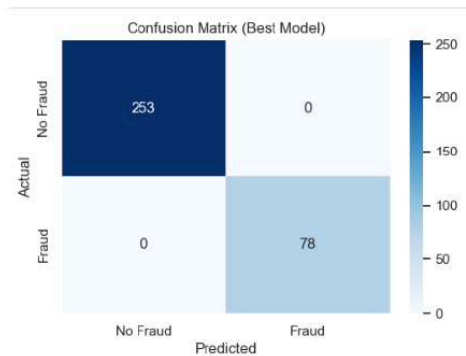
Precision: 1.0

Recall: 1.0

F1 Score: 1.0

Accuracy with Best Hyperparameters: 1.0

Confusion Metrics:



Sensitivity (Recall) - Best Model: 1.0
Specificity - Best Model: 1.0
Precision - Best Model: 1.0
Recall - Best Model: 1.0
F1 Score - Best Model: 1.0

Evaluation And Conclusion -

Evaluation of Logistic Regression Model

The Logistic Regression model, using an optimal cutoff of 0.5 , achieved the following performance metrics on the validation data:

- Accuracy: 0.8042
- Sensitivity (Recall for Class 1 0.5294
- Specificity (Recall for Class 0 0.8899
- Precision: 0.6000
- Recall: 0.5294
- F 1 Score: 0.5625|

The confusion matrix reveals:

- True Positives (TP): 18 (Fraudulent cases correctly identified)
- True Negatives (TN): 97 (Non fraudulent cases correctly identified)

- False Positives (FP): 12 (Non fraudulent cases incorrectly predicted as fraudulent)
- False Negatives (FN): 16 (Fraudulent cases incorrectly predicted as non fraudulent)

Evaluation of Random Forest Model

The Random Forest model, using the selected important features, demonstrated the following performance on the validation data:

- Accuracy: 0.7692
- Sensitivity (Recall): 0.2059
- Specificity: 0.9450
- Precision: 0.5385
- Recall: 0.2059
- F 1 Score: 0.2979

The confusion matrix reveals:

- True Positives (TP): 7
- True Negatives (TN): 103
- False Positives (FP): 6
- False Negatives (FN): 27

Conclusion

Comparing the performance of the two models on the validation data:

- Accuracy: The Logistic Regression model (0.8042) slightly outperformed the Random Forest model (0.7692) in terms of overall accuracy.
- Sensitivity (Recall): The Logistic Regression model (0.5294) showed a significantly higher ability to correctly identify fraudulent cases compared to the Random Forest model (0.2059).
- Specificity: The Random Forest model (0.9450) was better at correctly identifying non fraudulent cases than the Logistic Regression model (0.8899).
- Precision: The Logistic Regression model (0.6000) had a slightly higher precision, meaning that when it predicted a case as fraudulent, it was more likely to be correct than the Random Forest model (0.5385).
- F1 Score: The Logistic Regression model (0.5625) achieved a higher F1 score, indicating a better balance between precision and recall compared to the Random Forest model (0.2979).

In summary, based on these metrics on the validation data, the Logistic Regression model appears to be a better overall performer for this fraud detection task. It demonstrates a better balance between correctly identifying fraudulent cases (sensitivity/recall) and minimizing false positives (precision), leading to a higher F1. However, it's important to note the trade offs. The Random Forest model exhibited higher specificity, meaning it was better at avoiding false alarms (incorrectly flagging non fraudulent transactions). The choice of the "better" model might ultimately depend on the specific business context and the relative costs associated with false positives and false negatives. If minimizing false alarms is a higher priority, the Random Forest model might be considered despite its lower sensitivity. Conversely, if capturing as many fraudulent cases as possible is crucial, the Logistic Regression model seems more suitable.